

“Do Bike Lanes Reduce Congestion” Is the Wrong Question

Emma Durand-Wood · October 3, 2024



The logic by proponents of bike infrastructure goes like this: If more people use a bike rather than take a car, then there will be fewer cars on the road, causing less congestion.

On the other hand, folks who oppose bike lanes say that by taking space away from vehicles, the bike lanes actually cause congestion.

So which is it? I’ve been pondering this for some time, and I’ve come to the conclusion that asking whether bike lanes reduce congestion is the wrong question.

Congestion Is Not the Enemy

This might sound counter-intuitive, but bear with me: Most of us think of congestion as a scourge that must be eliminated, but there’s a different way to look at it.

Congestion is a byproduct of people getting to places they want and need to go. A recent [Transportation for America report](#) put it this

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Congestion is an issue to address, but it is also a symptom of success—a sign of a place people want to be. Focusing on congestion mitigation misses the bigger picture. We should focus instead on the goal of providing and improving access.

The core purpose of transportation infrastructure is to provide access: to work, education, healthcare, groceries, recreation, and all other daily needs. We should be prioritizing investments based on how well they connect people to jobs and services, not how much more quickly cars will be able to travel on a certain segment of highway.

I know this isn't a very satisfying assessment. Whether we're in our own car, in a cab or on a bus, being stuck in traffic drives us all crazy. However, the only real way to fight congestion is to make our transportation system more efficient. And we can do that by prioritizing the most efficient types of transportation — that is, the ones that take up the least amount of space.

When we argue that bike lanes will improve congestion, the vision that comes to mind is a perfect one-to-one conversion of cars to bikes. Fewer cars jamming up the road, free flowing travel within the bike lanes...problem solved! But that's an oversimplified prediction. Humans and their transportation choices are far more complex and unpredictable than that!

Ever heard of [traffic evaporation](#)? That's the term for why long-term chaos seldom arises when roads close or space is reallocated to other modes like walking, biking or public transit. People adapt in so many different ways: They change their routes, their destinations, their travel times, etc. They cope with the new reality, and they do not wind up stuck in a traffic apocalypse!

We can't accurately predict how people will adapt their travel plans or what environmental changes will occur in the surrounding area, and that's one of the reasons [traffic modeling](#) is a notoriously flawed and unreliable practice.

On the other hand, we do know one thing for certain thanks to ample research on induced demand: Adding capacity to roads doesn't solve



Because congestion is so nuanced, so slippery, so hard to pin down, framing bike infrastructure as primarily a tool to fight congestion misses the point. But there are many other compelling reasons to build bike lanes!

Bike Lanes Accomplish More Than One Thing at a Time

The most common objection I hear to bike lanes is that they will result in the loss of car travel lanes, therefore making things worse for folks behind the wheel. But this isn't always the case. Many successful bike lane projects preserve existing lanes, merely narrowing them to make room for the bike lanes.

On top of retaining these travel lanes, reducing their width brings significant safety benefits to the street.

Last year, researchers from the Johns Hopkins Bloomberg School of Public Health published some [incredible findings](#): Narrower vehicle lanes have significantly fewer non-intersection crashes than wider ones. (Check out this [Strong Towns podcast with the report's principal investigator](#), Shima Hamidi.)

While the goal may have simply been to accommodate the bike lane, this is a great way to do two things at once, creating both financial and safety benefits at the same time.

Bike Lanes Build Transportation Capacity

"The bike lanes are always empty, but the roads are bursting at the seams." We've all heard this one before, or thought it ourselves. But there's a reason those bike lanes look empty, and it's simple geometry: Bike lanes have much greater capacity than car lanes because bikes take up much, much less space than cars do.

well-used bike lane will appear underused compared to a car lane. They are just more efficient.

Do under-used bike lanes (i.e. ones that are legitimately empty) exist? It's possible. But that's more likely due to the lanes being poorly connected to other safe infrastructure.

If the goal is to increase the capacity of a street, there are a few ways to do this, and they involve prioritizing the most efficient types of travel, which are those that move the greatest number of people in the least amount of space: public transit, biking and walking.

Even if vehicle lanes are reduced (which isn't a foregone conclusion), an overall increase in capacity can be achieved with bike lanes, simply because they can accommodate so many more travelers in such a relatively small amount of space.

Bike Lanes Bring Customers to Businesses

You've probably heard it said that bike lanes are a death knell for small businesses, usually because of the loss of adjacent parking. But the opposite is true more often than not. Bikes are a boon for business, with plenty of research showing that customers who arrive by bike spend as much as those who arrive by car, with some studies even finding that [bike customers shop more often, spending more overall](#) in a given commercial area.

If it's deemed important to retain parking, this can be integrated into the new street layout with parking-protected bike lanes, for instance.

Rachel Quednau's article "[How Bike Lanes Benefit Businesses](#)" includes lots of examples from cities across North America.

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Building and maintaining transportation infrastructure is generally one of a town or city's biggest line items. Bike lanes are cheaper to build, cost (orders) less to maintain and, as mentioned above, are good for business (which means good for city coffers).

Maybe you've heard this famous story about Copenhagen: The city was on the verge of bankruptcy, but, wanting to promote development, opted to invest in the cheapest possible infrastructure: bicycle infrastructure. As former city councilor Morten Kabell [put it](#), "When mayors in other cities ask me how Copenhagen afforded to invest in its cycling networks, I ask them how on earth they have been able to afford highway projects. ... We invested in bike lanes because that was the cheapest option."

That decision helped make Copenhagen a place where 62% of citizens use a bike to get to school or work every day. And their bottom line is better for it: Those infrastructure projects pay for themselves in spades. Kabell describes one cycling/walking bridge as paying for itself in under seven years.

Safer Streets for Everyone, Including Drivers

Brand-new research from Wesley Marshall and Nicholas Ferenchak found that [bicycling activity is significantly associated with better safety for all road users](#). In other words, towns and cities where a lot of people bike tend to have better safety outcomes for drivers and pedestrians.

Part of this is the safety in numbers factor — the more cyclists there are, the more visible and expected they are to drivers. But the physical factors that make biking safe and comfortable play a key role, too. Good bike infrastructure does double duty: It not only protects folks on bikes but also serves as a form of traffic calming

environments.

As David Zipper, a senior fellow with MIT Mobility Initiative, [put it](#), “Bicyclists are an indicator species for general street safety (among drivers, pedestrians, etc). If you see lots of people cycling, city streets are probably safe. If not, you've got a problem.”

Bike Lanes Are an Investment in the Future

When a town or city opts to install [safe bike infrastructure](#), it's not just about giving existing bike riders a safer route. It's usually also about encouraging more people to take a trip by bike. In other words, bike lanes aren't just about the present, they're about the future and what types of travel behavior the city is trying to incentivize. In that way, building bike infrastructure is a way of inducing demand for nonvehicular travel.

That image of a driver stuck in traffic seeing people zipping by in bike lanes, thinking “I should bike next time” may be oversimplifying, but the principle stands. Give people a more efficient way to travel and some will take you up on your offer.

When we make fighting congestion our priority above all else, we are missing the point of what travel is: Getting to and from the places we live, work and play. Fighting congestion is about addressing symptoms (or, put another way, trying to improve transportation for its own sake), while adding bike lanes is about providing access for more people, which is the goal of transportation to begin with.

Cities are complex places, and our care, attention and focus should be directed toward addressing financial and environmental sustainability, liveability, and prosperity for all. Our infrastructure choices, especially in how we design our streets and roads, have an [enormous influence on these qualities](#). In my books, of all the



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